Inch-Pound

MIL-T-58078B(AV)
11 August 1989
SUPERSEDING
MIL-T-58078A
25 February 1985
MIL-T-58078
16 July 1964

MILITARY SPECIFICATION

TESTER, PITOT AND STATIC SYSTEMS, DIGITAL

This specification is approved for use by US Army Aviation Systems Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification establishes the minimum design performances and test requirements for a portable pitot and static tester to be used for the inspection and acceptance of pitot and static systems including altimeters, vertical velocity indicators, air speed indicators and air data computer systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Aviation Systems Command, ATTN: AMSAV-EIG, 4300 Goodfellow Boulevard, St. Louis, MO 63120-1798, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4920

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SPECIFICATIONS

FEDERAL

PPP-B-576 Box Wood Cleated, Veneer, Paper Overlaid

MILITARY

MIL-T-704 Treatment and Painting of Materiel
MIL-T-28800 Test Equipment for Use with Electrical and Electronic Equipment, General Specification for
MIL-C-46168 Coating, Aliphatic Polyurethane, Chemical Agent Resistant

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage
MIL-STD-859 Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MIL-STD-2073 DOD Material Procedure for Development and Application of Packaging Requirements

AIR FORCE-NAVY AERONAUTICAL

AN 737 Clamp, Hose, Aircraft
AN 815 Union-Flared Tube
AN 919 Reducer-External Thread Flared Tube
AN 6270 Hose Assembly Swivel Fitting, Low Pressure

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitations.

DRAWINGS

USAAVSCOM

SK-ED-0017 Flush Static and Pitot Adapter
SK-ED-0018 Pitot Head Adapter

PUBLICATIONS

NASA TN D-822 Tables of Airspeed, Altitude, and Mach Number Based on Latest International Values for Atmospheric Properties and Physical Constants

U.S. Standard Atmosphere, 1976

(Copies of Specifications, standards, drawings, publications and other Government documents required by contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issue of the documents not listed in the DODISS are the issue of the document cited in the solicitation (see 6.2).

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS
IEEE-STD-488-1978 Standard Digital interface for Programmable Instrumentation

(Application for copies should be addressed to the IEEE Standard Sales, 345 East 47th Street, New York NY 10017)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets of MS Standards), the text of this specification shall take precedence. Nothing in this specification; however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The test set shall be designed for functional operation and use at unit (flight line) and intermediate (shop) levels of maintenance. The primary function of the test set will be to provide regulated total and static pressure outputs simulating altitudes and airspeeds for evaluation of aircraft's air data systems and associated equipment. The test set shall consist of a main pneumatic/electronic assembly, a combination case, a remote unit (see 3.4.4)., a power cable assembly (see 3.4.6.1), pneumatic hose assemblies (see 3.4.7.2), and other accessories and connections.

3.1.1 First Article. When specified in the contract or purchase order, a sample will be subjected to first article inspection (see 6.3).

3.1.2 Government Furnished Property. There shall be no requirement for Government furnished property.

3.2 Characteristics. Unless otherwise specified, the test set shall meet the requirements of MIL-T-28800 for Type II, Class 2, Style C, and Color P.

3.2.1 Performance Characteristics. The test set shall provide two primary pressure outputs: simulated altitude (see 6.4) and simulated airspeed (see 6.4) The test set shall be microprocessor controlled unit to incorporate self test (see 3.5.3) to facilitate its operation.

3.2.2 Physical Characteristics. The test set shall not exceed 3.5 cubic feet in volume, when housed in the combination case with the cover closed. No external dimension shall exceed 25 inches. The weight of the complete test set including the main unit, remote unit, adapters, hoses, cover and accessories shall not exceed 65 pounds and shall be kept to a minimum. The weight of the remote unit shall not exceed 10 pounds.

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3.2.3 Reliability. The reliability requirements shall be as specified in MIL-T-28800, except the mean time between failure (MTBF) shall be 1,000 hours and the lower test MTBF shall be 500 hours.

3.2.4 Maintainability. The maintainability requirement shall be as specified in MIL-T-28800, except the mean time to repair (MTTR) shall be one hour and the maximum corrective maintenance down time shall be three hours.

3.2.5 Durability. The systems shall have a minimum useful service life of 3000 hours before wearout failure occurs.

3.2.6 Environmental Requirements. The test set shall conform to the environmental requirements of MIL-T-28800 for Type II, Class 2, Style C.

3.2.6.1 Acoustic Noise. The acoustic noise requirement shall be as specified in MIL-T-28800.

3.2.6.2 Operating Altitude. The test set shall meet the performance and accuracy requirements specified herein when operated in altitudes from -1000 to 15,000 feet. Changes in the ambient barometric pressure shall not affect the performance and accuracy requirements of the test set.

3.2.7 Scheduled Maintenance. The test set shall not require scheduled maintenance.

3.2.8 Calibration Interval. The minimum off-line calibration interval shall be 12 months, with a 0.85 probability that all components requiring calibration shall be within tolerance at the end of this period.

3.2.8.1 Calibration Equipment. New calibration standards shall not be required for calibration of the test set. Calibration standards shall be selected from equipment in the Government inventory for this purpose.

3.2.9 Support Equipment Selection. Peculiar support equipment (see 6.4) shall not be required for maintenance of the test set. Support equipment selection shall be based on equipment already existing in the Government inventory, such as general purpose electronic test equipment.

3.3 Design and Construction. The test set shall be a single, integral unit, self-contained within its case. The case shall serve to protect the complete test set and shall have a compartment in the cover for storage of all accessories including the remote unit and cable.

3.3.1 Attitude. The test set shall be operable in a horizontal or vertical position and shall be self supported on either position.

3.3.2 Parts, Materials and Processes. Parts, materials, and processes shall be in accordance with MIL-T-28800.

3.3.3 Electromagnetic Interference (EMI). The test set shall conform with the requirements of MIL-T-28800.

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3.3.4 Finish. External surface shall be treated per MIL-T-704 Type G (non-Ferrous) or Type F (Ferrous). Finish with two coats of 383 Green per MIL-C-46168.

3.3.5 Human Engineering. The human engineering requirements shall be in accordance with MIL-T-28800.

3.4 Design Requirements.

3.4.1 Electrical Power. The test set shall operate from any of the following power sources:

28 volt direct current (dc), +4 to -6 volts
115 volt ± 10 percent, 50 to 400 Hertz single-phase

3.4.1.1. Input Power Switch. The input power switch shall be as specified in MIL-T-28800. The power switch shall function as an emergency power turn-off. Turning the switch to the off position at any moment during operation of the test set shall not damage the test set and the unit under test. After turn-off the test set shall automatically vent all pneumatic outputs to the ambient barometric pressure at a controlled rate. A power on indicator shall indicate whenever power is applied to the test set. An input power switch shall be located on the front panel of the set and on the remote control unit.

3.4.1.2 Overload Protection. The test set shall provide protection against overload conditions and incorrect application of input power.

3.4.1.3 Transient State Power Conditions. The operation of the test set shall not be impaired by the transient state power condition specified in MIL-T-28800 for Type II equipment.

3.4.2 External Connections. Electrical and pneumatic connections shall be made on the front panel. The electrical cables and pneumatic hoses, when connected to the test set, shall not interfere with the viewing and operation of the test set when used in either a horizontal or vertical attitude.

3.4.2.1 Altitude Encoder Connection. The test set shall have a connector for connecting the encoder output of encoding altimeters to the test set. An adapter cable for connecting the AAU-32 encoder output to the test set shall be provided.

3.4.3 Operation Indicators, Displays and Controls. The test set shall provide displays, controls and status indications to the user. The test set shall have as a minimum, but not limited to the following indicators, displays and controls:

Elapsed time indicator	see 3.4.3.1
Power on indicator and switch	see 3.4.1.1
Self test fault indicators, selector	see 3.5.3
Simulated altitude display, selector, indicator	see 3.5.1
Simulated airspeed display, selector, indicator	see 3.5.1
Test set ready indicator	see 3.4.3.2
Decoded altitude display	see 3.5.1.4
Altitude code bit display	see 3.5.1.5

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Auto-exercise indicator	see 3.5.5
Leakage test mode indicator, selector	see 3.5.2
Altitude slew rate display, selector, indicator	see 3.5.1
Airspeed slew rate display, selector, indicator	see 3.5.1
Ambient barometric pressure indicator, vent switch	see 3.5.10.8
Indicated/true airspeed control	see 3.5.1.2

3.4.3.1 Elapsed Time Indicator. The elapsed time indicator shall indicate the total accumulated time in hours the test set has been operating (see 6.5).

3.4.3.2 Test Set Ready Indicator. The test set ready indicator shall indicate when the test set is ready to accept an input from the user.

3.4.4 Remote Unit. A hand held remote unit shall provide the primary means of controlling the test set. The remote unit shall be compact, lightweight and portable unit, housed in a case to protect it from the environmental and EMI conditions. The environmental and EMI requirements for the remote control units are the same requirements that apply to the test set. The weight of the remote unit and cable assembly shall not exceed 10 pounds.

3.4.4.1 Controls, Displays, Indicators. All controls, displays and indicators required to operate the test set shall be included in the remote unit. The remote unit shall only contain the circuits necessary to interface with the main pneumatic/electronic assembly (main unit). The only connection between the remote unit and the main unit shall be a single 20 foot long, electrical cable assembly with a standard MS connector at both ends.

3.4.4.2 Remote Unit Operation. The test set may not be required to be operated manually with the remote unit disconnected. The IEEE-488 control bus shall operate with the remote unit connected and disconnected.

3.4.4.3 Keyboard. A general numeric keyboard shall be provided as the primary input device for the operators to enter test points, rates of change, etc.

3.4.4.4 Remote Unit Bracket. A bracket or other means of securing the remote control unit to the front panel of the test set shall be provided.

3.4.5 Pressure-Vacuum Pump. The pressure-vacuum pump shall be a motor driven dry type pump. The pump assembly shall incorporate an elapsed time indicator to indicate the number of hours the pump has been operating (see 6.5).

3.4.6 Internal Pneumatic Tubing. The internal tubing of the test set shall be designed for ease of maintenance. Flexible hoses shall be used.

3.4.7 ACCESSORIES

3.4.7.1 Power Cables. Power cables, extensions, and adapters shall conveniently connect the tester to any of the power sources listed in 3.4.1. Power cables shall possess adequate grounding and shall be at least 14 feet in length.

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3.4.7.2 Hose and Connections. Hoses, clamps, couplings, sleeves, and reducers, as applicable, shall connect the tester to points in the aircraft plumbing, to the pitot pressure port located in the aircraft pitot head, and to the static pressure port which may protrude, or be flush located, on the outer surface of the aircraft. Adapters shall also be provided to connect the tester directly to the instrument cases. For bench-testing uninstalled instruments, valves shall be provided with the accessory hose connections to isolate the instruments being tested for leakage. Acceptable hose and fittings with their applicable standards or drawings, are listed in Table I. Adapters shall be in accordance with Drawings SK-ED-0017 and SK-ED-0018.

TABLE I

External Hose and Connections

Quantity	Nomenclauture	Standards or Drawings
1	Hose Assembly 5 ft. (Lg)	AN6270-4
2	Hose assembly 6 ft. (Lg)	AN6270-4
2	Reducer, 1/4 in. to 3/16 in.	AN919-2D
1	Reducer, 1/4 in. to 1/8 in.	AN919-1D
2	Union, 1/4 in. flared tube	AN815-4D
1	Union, 3/16 in. flared tube	AN815-3D
1	Clamp, sleeve	AN737TW34-38
1	Pitot head adapter	DWG SK-ED-0018
1	Flush static port adapter	DWG SK-ED-0017

3.5 Performance Requirements.

3.5.1 Altitude and Airspeed Simulation. As a minimum, the test set shall simulate and display any selected altitude and airspeed condition continuously within the limits specified herein. The pressure regulation system of the test set shall not be affected by changes in the ambient barometric pressure. The altitude pressure output shall be based on a fixed barometric pressure reference of 29.9213 inches Hg, and shall be calibrated to the standard atmosphere as listed in MIL-STD-859 and in the U.S. Standard Atmosphere, 1976. The airspeed pressure output shall be calibrated to NASA TN D-822. The altitude and airspeed displays shall provide for both commanded and measured values. All other commanded and displayed values shall be displayed whenever selected by the operator. The altitude and airspeed rate displays shall show commanded values and an indicator/display shall be provided to let the operator know when the measured rate is equal to the commanded rate within specified tolerances.

3.5.1.1 Ranges, Resolution and Accuracies. The test set shall simulate pressure/vacuums in accordance with the ranges, resolution, units and accuracies of Table II. The test set shall have the capability to display in psi pressure units.

3.5.1.2 Simulated, True Airspeed Control. The test set shall simulate indicated airspeed and true airspeed. A control shall be provided to select the desired kind.

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TABLE II
RANGES, RESOLUTIONS AND ACCURACIES

PARAMETER	COMMAND/DISPLAY/OUTPUT
ALTITUDE	-1500 to 35,000 ft
Resolution	1 ft
Accuracy	+/-10 feet or +/-0.1% of cmd
ALTITUDE RATE (RATE OF CLIMB)	
Range	0 to ± 10,000 ft/min
Resolution	100 ft/min
Accuracy	+/-2 fpm or +/-2% of cmd
DECODED ALTITUDE	
Range	-15000 to 35000 ft
Resolution	100 foot increment
AIRSPEED	
Range	20 to 400 knots
Resolution	0.1 knots
Accuracy	1.5 knots from 20 to 60 knots +/-1.0 knots above 60 knots
AIRSPEED RATE	
Range	0 to 400 knots
Resolution	100 knots/min
Accuracy	+/-2 knots/min or +/-2% of cmd

3.5.1.3 Display Characteristics. All displays shall be digital and sunlight readable. The displays shall have a minimum contract ratio of 3 over a field of view of ± 45 degree cone with a target contrast ratio of 45 over a ± 60 degree cone. Displays shall be as follows:

DISPLAY	NUMBER OF DIGITS (minimum)	CHARACTER HEIGHT (minimum)
Altitude	5	0.375 inches
Airspeed	5	0.375 inches
Altitude rate	4	0.250 inches
Airspeed rate	3	0.250 inches

3.5.1.4 Decoded Altitude. The test set shall display the output of altitude reporting encoders of altimeters or equipment which incorporate encoders for

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automatic altitude reporting. The display shall be in feet of altitude and 100 foot steps.

3.5.1.5. Code Bit Displays. A 10-bit display shall be available to indicate the status of each bit of an altitude reporting encoder. The display shall be for trouble shooting those units that have an erroneous digital reading to determine which code bits are missing.

3.5.2 Leakage Test Mode. On this mode the test set shall measure the leakage on the static and dynamic pneumatic outputs and display the corresponding leakage on the simulated altitude and airspeed displays in units of feet per minute and knots per minute, respectively. The test set shall be capable of performing individual and simultaneous pneumatic channel leak tests. The test set shall perform this test regardless of the starting conditions of simulated altitude and airspeed with an accuracy of 50 feet per minute on the altitude channel and one knot per minute on the airspeed channel. The leakage test mode indicator shall indicate that the test set is in this mode of operation until the mode is deactivated by the user.

3.5.3 Self Test. The test set shall execute the self test routine when power is applied to the test set (turn-on). The self test routine shall also be initiated manually by a switch in the front panel. The self test routine shall also be executable through the IEEE-488 control bus. Adequate status information or flags shall be available through the bus in case a malfunction condition exists. When a lamp test is initiated, all lamps and other indicating devices shall illuminate. The lamp test shall include the testing of the numeric displays. A malfunction in the self test circuits shall not impair the operation of the test set. If a malfunction is detected by the self test, the test set shall vent the pneumatic outputs to the atmospheric pressure at the controlled rate of 5,000 feet per minute and 100 knots per minute, respectively. The self test sequence shall not require intervention of the user for its completion. The self test routine shall, as a minimum, test the following functions:

a. The static and dynamic pressure sensors.

b. The pressure generating system.

c. The pressure controlling system.

d. Lamp test (includes numeric displays).

3.5.4 Self Test fault Isolation Criteria. The self test routine shall detect faults and automatically isolate the fault to a shop replaceable unit (SRU) ambiguity group to meet the requirements of Table III. When a fault is detected the test set shall indicate the faulty SRU or a group of SRUs.

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TABLE III SELF TEST SRU FAULT ISOLATION AMBIGUITY GROUP SIZE.

Percentage of faults isolated	SRU ambiguity group size	Time limit (minutes)
100%	3 SRU's	5
95%	2 SRU's	10
90%	1 SRU's	15

3.5.5 Auto-Exercise Function. The test set shall provide for an automatic function to exercise the complete pneumatic system throughout the entire pneumatic range. The auto-exercise (see 6.4) function shall be operator initiated and an indicator shall indicate that the test set is executing the auto-exercise routine. The auto-exercise routine shall consist of cycling three times through the entire pneumatic pressure range. The test set shall complete the entire auto-exercise routine in 15 minutes or less. During the auto-exercise function the test set shall isolate the pneumatic outputs and display the simulated altitudes and airspeeds. This function may not be required if the contractor selects pneumatic sensors that do not require this type of excitation to achieve the required accuracies.

3.5.6 External Pressure Source. The test set shall be capable of receiving pneumatic vacuums and pressures from an external pneumatic pressure source. The pressure input fittings shall be accessible by removing a cover from an opening in the front panel. A switch shall be provided inside the opening for turning off the pneumatic generating systems in the test set. Connection of an external source shall be completed by connecting no more than two hoses and by operating no more than two pneumatic valves.

3.5.7 IEEE-488 Bus Interface. The test set shall be capable of interfacing with an IEEE-488 control bus in accordance with IEEE-STD-488-1978 and as specified in MIL-T-28800. The test set shall be controlled through the IEEE-488 control bus by using an external bus controller. All functions, indications and readings on the test set shall be available through the control bus. The IEEE-488 bus address switch shall be located inside the access panel for the external pressure source (see 3.5.6). It is acceptable to operate the remote unit from the IEEE-488 bus.

3.5.8 External Volume Range. The test set shall operate and perform as specified herein with any combination of external volumes (see 6.4) connected to the pneumatic outputs. The external volume range is:

a. Connected to the static output, from 5 to 250 cubic inches.

b. Connected to the dynamic output, from 5 to 100 cubic inches.

3.5.9 Stability. The test set shall maintain stable pressure control and regulation of simulated altitude and airspeed, while operating into any combination of external volumes as specified herein.

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3.5.9.1 Altitude Stability. The maximum allowable variation of simulated altitude shall be ±3 feet or ±0.02% of simulated altitude, whichever is greater.

3.5.9.2 Airspeed Stability. The maximum allowable variation of simulated airspeed shall be ±1 knot throughout the entire simulated airspeed range.

3.5.9.3 Oscillations Within Stability Limits. The test set shall not produce rapid oscillations, surges or transients within the stability limits specified herein. Any oscillations shall be exhibited as a slow drift. The duration of a drift from a maximum to a minimum stability limit shall not exceed one minute.

3.5.9.4 Stability of Pressure Interaction. The test set shall have minimum interaction between the simulated altitude output and the simulated airspeed output as follows:

a. At indicator readings of 5,000 feet altitude and 50 knots airspeed, an increase in airspeed to 300 knots shall not produce an altitude change in excess of ± 50 feet.

b. At indicator readings of 2,000 feet altitude and 50 knots airspeed, an increase in altitude to 35,000 feet shall not produce an airspeed change in excess of ±2 knots.

3.5.10 Protection to the Unit Under Test. The test set shall provide protection to the unit under test and to the test set. The test set shall prevent pressure surges, pressure transients, over-pressure, and under-pressure in both, the test set and the unit under test. The test set shall provide pressure balance protection to the altimeter and airspeed indicator.

3.5.10.1 Overshoot. Overshoot regulation shall be accomplished by automatically reducing the commanded slew rate (altitude or airspeed) when the simulated pressure at the output approaches the pressure equivalent to the commanded altitude or airspeed.

3.5.10.2 Negative Differential Pressure. The test set shall limit the negative differential pressure from exceeding 0.5 inches Hg.

3.5.10.3 Static Pressure Limiting. The test set shall maintain the static pressure between a maximum of 32 inches Hg and a minimum of 6.874 inch Hg.

3.5.10.4 Total Pressure Limiting. The test set shall not permit the total pressure to exceed the static pressure by more than 8.866 inches Hg.

3.5.10.5 Pneumatic Filtering. The test set shall provide pneumatic filtering to preclude the transmission of foreign matter to the unit under test and to provide protection to the test set.

3.5.10.6 Water Extraction. The test set shall extract the water that may accumulate in the internal pneumatic system of the test set. Water shall be extracted when the test set is operated on either a horizontal or a vertical attitude and removal of the unit from its enclosure shall not be required. Water extraction shall be an automatic function of the test set.

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3.5.10.7 Human Error Design Protection. Front panel control functions shall be designed so that neither incorrect adjustment nor random sequencing of functions cause damage to the test set and the unit under test.

3.5.10.8 Pneumatic Output Venting. The ambient barometric pressure indicator shall indicate when the pneumatic outputs of the test set are at the ambient barometric pressure. When the ambient vent switch is activated the test set shall automatically vent the pneumatic outputs to the ambient barometric pressure at a controlled rate of 5,000 feet per minute and 100 knots per minute, respectively. The ambient vent condition may be considered as a "reset function."

3.5.11 Test Set Leakage. The test set's leakage shall not exceed 50 feet per minute and one knot per minute on the static and dynamic outputs under the conditions of 35,000 feet and 400 knots simulated altitude and airspeed, respectively.

3.6 Warm-up. Warm-up shall be as specified in MIL-T-28800, except that the warm-up period shall not exceed 15 minutes.

3.7 Workmanship. The workmanship requirements shall be as specified in MIL-T-28800.

3.8 System Safety. The test set shall comply with safety requirements of MIL-T-28800 for Type I equipment.

4 QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspections as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. All items shall conform to all requirements of Sections 3.0 and 5.0. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements for the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of Inspections. The inspection requirements specified herein are classified as in (a) and (b):

a. First article inspection .....See 4.3.

b. Quality conformance inspection .....See 4.4.

4.3 First Article Inspection. First article inspection shall be as specified in MIL-T-28800 and shall include the tests specified in Table IV.

4.4 Quality Conformance Inspection. The quality conformance inspection shall be as specified in MIL-T-28800 and shall include the tests specified in Table IV.

4.4.1 Optional Quality Assurance Testing. The procuring activity may substitute a contractor's approved quality assurance system for any or all tests specified in Table IV in accordance with MIL-T-28800.

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TABLE IV. TEST GROUPS AND TEST METHODS.

Description	Requirement	Test Method
Group A:		
Preoperational inspection:		
Mechanical and visual examination	3.0	4.5.2.1
Electrical circuit configuration	3.4.1	4.5.2.1
Random Vibration and Temperature Cycling	3.7	4.5.2.1
Level A performance tests:		
Open circuit voltage	3.3.5	4.5.2.2
Leakage current	3.3.5	4.5.2.2
Level A performance evaluation	3.3.5	4.5.2.2
Group B:		
Altitude and airspeed simulation	3.5.1	4.5.3
Slew rates tests	3.5.1	4.5.3
Leakage test	3.5.2	4.5.3
UUT protection	3.5.10	4.5.3
Group C		
Temperature-humidity tests	3.2.6	4.5.4
Altitude	3.2.6	4.5.4
Vibration	3.2.6	4.5.4
Mechanical shock tests	3.2.6	4.5.4
Water resistance tests	3.2.6	4.5.4
Electrical power tests	3.4.1	4.5.4
Input power	3.4.1	4.5.4
Voltage and frequency variation	3.4.1	4.5.4
Frequency transient	3.4.1	4.5.4
Group C verification	3.2.6	4.5.4
Group D:		
Salt atmosphere tests	3.2.6	4.5.5
Explosive atmosphere tests	3.2.6	4.5.5
Sand and dust tears	3.2.6	4.5.5
Electromagnetic interference tests	3.3.3	4.5.5
Groups D verification	3.2.6	4.5.5
Group E:		
Maintainability	3.2.4	4.5.6
Self test demonstration	3.5.3	4.5.6
Acoustic noise tests	3.2.6.1	4.5.6
Dimensions and weight	3.2.2	4.5.6
Mechanical stability tests	3.3.1	4.5.6
Packaging inspection	5.1	4.5.6
Group F		
Reliability	3.2.3	4.5.7

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4.5 Examination and Test Methods.

4.5.1 Satisfactory Operation . The satisfactory operation check (see 6.4) and satisfactory operation test (see 6.4) shall, as a minimum, test the following criteria:

a. Satisfactory operation check. Whenever a satisfactory operation check is required, a self test routine (see 3.5.4) shall be performed. Satisfactory completion of the self test routine shall be considered a satisfactory operation check.

b. Satisfactory operation test. The following characteristics shall be verified by the satisfactory operation test:

(1) Altitude simulation range.

(2) Airspeed simulation range.

(3) Test set leakage and leakage test mode.

(4) Autoexercise function.

(5) Altitude slew rate.

(6) Airspeed slew rate.

(7) Negative differential pressure.

(8) Remote control operation.

4.5.1 Group A Inspection. Group A inspection shall consist of the preoperational inspection of 4.5.2.1 and the Level A performance tests of 4.5.2.2.

4.5.2.1 Preoperational Inspection. The preoperational inspection requirement shall be as specified in MIL-T-28800, Type III.

4.5.2.2. Level A performance test. The level A performance tests shall be as specified in MIL-T-28800 and in (a) and (b):

a. Leakage current test. A leakage current test shall be performed in accordance with MIL-T-28800.

b. Level A performance evaluation. Level A performance evaluation shall be conducted to ensure that all functions and modes of operation of the test set comply with the requirements specified herein. Particular attention shall be given to the following characteristics:

(1) Altitude simulation.
(2) Airspeed simulation.
(3) Test set leakage.
(4) Autoexercise function.

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(5) Altitude slew rate.
(6) Airspeed slew rate.
(7) Negative differential pressure.
(8) Remote control operation.

4.5.3 Group B Tests. The Group B tests consist of the Level B performance tests and shall be performed as specified in MIL-T-28800. The level B performance tests shall, as a minimum, test the performance characteristics of (a) through (e).

a. Altitude and airspeed simulation tests.
altitude simulation range and accuracy
airspeed simulation range and accuracy
altitude and airspeed maximum limits
altitude and airspeed stability
external volume range

b. Altitude and airspeed slew rate tests.

c. Leakage test mode and test set leakage tests.

d. UUT protection
overshoot
negative differential pressure
pneumatic filters and air quality
water extraction

4.5.4 Group C Test. The Group C tests shall be as specified in MIL-T-28800 and in (a) and (b):

a. Temperature-humidity tests. The temperature-humidity tests shall be performed in accordance with MIl-T-28800 for class 2 equipment. The maximum allowable degradation from the test set's performance requirements when subjected to high and low temperature extremes of the temperature-humidity ranges shall not exceed the following limits:

(1) High temperature (50 to 71 degrees centigrade.) The simulated altitude accuracy shall be ±20 feet or ±0.2%. whichever is greater, through the range of -1,500 to 35,000 feet. The simulated airspeed accuracy shall be ±3 knots throughout the entire simulated airspeed range.

(2) Low temperature (-10 to -40 degrees centigrade.) The simulated altitude accuracy shall be +30 feet or ±0.25%, whichever is greater throughout the range of -1,500 to 35,000 feet. The simulated airspeed accuracy shall be ±4 knots through the entire simulated airspeed range.

b. Altitude test. The altitude tests shall be as specified in MIL-T-28800.

4.5.5 Group D Tests. The Group D tests shall be as specified in MIL-T-28800.

4.5.6 Group E Tests. The Group E tests shall be as specified in MIL-T-28800 and in (a) through (c).

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a. Maintainability demonstration tests. A maintainability demonstration test shall be conducted in accordance with MIL-T-28800.

b. Self test demonstration. A self test demonstration shall be conducted to demonstrate that the self test requirements of 3.5.3 are met.

c. Acoustic noise tests. The acoustic noise test shall be in accordance with MIL-T-28800.

4.5.7 Group F Tests. The Group F tests shall be performed in accordance with MIl-T-28800.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. Tester shall be packaged at level A or C, as specified (see 6.2).

5.1.1 Level A. Tester shall be packaged in a combination case with cover to meet level A requirement in accordance with MIL-STD-2073.

5.1.2 Level C. Tester shall be packaged in a manner to prevent deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be at level A. or C as specified (see 6.2.1).

5.2.1 Level A. Tester packaged as specified in 5.1.1 to meet level A requirements, shall be packed in exterior shipping container conforming to PPP-B-576, Style B, Class 2.

5.2.2 Level C. Tester packaged as specified in 5.1.1 or 5.1.2 shall be packed in a manner to ensure carrier acceptance and safe delivery to destination.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclaure shall be:

Tester, Pitot and Static Systems, Digital

6.0 NOTES.

6.1 Intended Use. The test set shall provide two primary outputs, static pressure and total pressure or pitot pressure. The differential pressure, defined as the pressure caused by the movement of the aircraft through the atmosphere, is given by the difference of total pressure and static pressure. Altitude is a function of static pressure and airspeed is a function of differential pressure. The test set will be used primarily at the unit (flight line) and intermediate (field shop) levels of maintenance. Its primary use will be to perform functional testing of air data systems, altimeters, and other associated avionics equipment.

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6.2 Ordering Data.

6.2.1 Acquisition Requirement. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Deviations, additions, substitutions, or deletions from the provisions of this specification.

c. Number of preproduction units to be submitted for first article testing, if other than specified in 4.3.

d. Levels of packaging and packing.

6.3 First Article. When a first article inspection is required, the items should be preproduction items.

6.4 Definitions and Abbreviations. For the purpose of this specification, the following definitions and abbreviations shall apply.

6.4.1 Auto-exercise. Automatic execution of a routine that cycles the pneumatic system (pump, transducer, etc) through the entire pneumatic pressure range to stabilize the pneumatic system.

6.4.3 Dynamic Output. The output of the test set which provides the dynamic pressure.

6.4.4 External volume. That closed volume connected to the output of the test set.

6.4.5 Indicated Airspeed. Actual instrument indication for a given flight condition. Factors such as altitude, instruments errors, and installation errors are not taken in consideration.

6.4.6 Peculiar Support Equipment. Support equipment specially developed for support of the test set.

6.4.7 Satisfactory Operation Check. A satisfactory operation check is a condition which employs a maximum number of active circuits of the test set with a minimum requirement for external input or evaluation equipment.

6.4.8 Satisfactory Operation Test. Satisfactory operation test requirements prove the continued satisfactory operation of the test set.

6.4.9 Self Test. A self test is a test or series of tests, performed by a device upon itself, which shows whether or not it is operating within designed limits.

6.4.10 Simulated Airspeed. The airspeed, in knots, simulated by the test set and corresponding to the true airspeed or indicated airspeed.

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6.4.11 Simulated Altitude. The altitude, in feet, simulated by the test set and corresponding to pressure altitude.

6.4.12 Slew Rate. The rate at which the test set increases or decreases the pneumatic outputs to attain a commanded altitude or airspeed.

6.4.13 Static Output. The output of the test set which provides the static pressure.

6.4.14 Static Pressure. Pneumatic pressure corresponding to an altitude.

6.4.15 True Airspeed. Free stream air velocity.

6.5 Elapsed time indicators. The test set shall have two elapsed time indicators, the test set indicator shall be visible from the outside and shall indicate the elapsed time the test set has been operating. The pneumatic pump indicator shall be permanently attached to the frame of the pump and shall indicate the elapsed time the pump has been operating.

6.6 Subject Term (Key word) Listing:

Test set
Pressure, dynamic
Hose, pneumatic
Total pressure, regulated
Static pressure
Altitude, simulated
Auto-exercise
Airspeed, simulated
Slew rate, altitude
Slew rate, airspeed
Output, static
Output, dynamic
Pneumatic outputs

6.7 Application of MIL-T-28800. (Cautionary note) This detailed specification supplements the general requirements specified in MIL-T-28800 for the type, class, style, and color specified herein. Classification by type, class and style automatically invoke requirements in MIL-T-28800. Bidders and contractors are cautioned to evaluate the applicable requirements of MIL-T-28800 and this document.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-58078B(AV)	Tester, Pitot and Static Systems, Digital FSC 4920

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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